

NEET Chemistry — Mole Concept

Ultra Premium Practice Sheet (20 Questions • High Level)

Q1. 1 mole of a gas occupies 22.4 L at STP. Volume at 2 atm and 273 K is:

- A. 22.4 L
- B. 11.2 L
- C. 44.8 L
- D. 5.6 L

Q2. Number of molecules in 44 g CO₂ is:

- A. 6.02×10^{23}
- B. 12.04×10^{23}
- C. 3.01×10^{23}
- D. 24.08×10^{23}

Q3. Empirical formula of compound with %C=40, %H=6.67, %O=53.33 is:

- A. CH₂O
- B. C₂H₂O₂
- C. CH₂O
- D. C₂H₄O

Q4. Assertion: Mole is unit of amount of substance. Reason: It represents fixed number of entities.

- A. Both correct explain
- B. Both correct not explain
- C. A correct R wrong
- D. Both wrong

Q5. 10 g CaCO₃ on heating gives CO₂. Volume at STP:

- A. 2.24 L
- B. 1.12 L
- C. 4.48 L
- D. 0.224 L

Q6. Limiting reagent is determined by:

- A. Moles
- B. Mass
- C. Volume

D. Density

Q7. Mass of 0.5 mole NaCl:

A. 29.25 g

B. 58.5 g

C. 11.7 g

D. 5.85 g

Q8. Molarity of solution with 1 mole solute in 2 L:

A. 0.5 M

B. 1 M

C. 2 M

D. 0.25 M

Q9. Number of atoms in 18 g H₂O:

A. 6.02×10^{23}

B. $3 \times 6.02 \times 10^{23}$

C. $2 \times 6.02 \times 10^{23}$

D. 1×10^{23}

Q10. Which has max molecules?

A. 1 g H₂

B. 1 g O₂

C. 1 g CO₂

D. 1 g N₂

Q11. Normality depends on:

A. Equivalent factor

B. Volume

C. Mass

D. Density

Q12. Equivalent weight of H₂SO₄:

A. 49

B. 98

C. 32

D. 64

Q13. Gas law $PV=nRT$, R depends on:

- A. Units
- B. Pressure
- C. Temperature
- D. Volume

Q14. Density of gas at STP:

- A. $M/22.4$
- B. $22.4/M$
- C. $M \times 22.4$
- D. None

Q15. Which is limiting reagent in reaction?

- A. Less moles
- B. Less mass
- C. Less volume
- D. None

Q16. Avogadro number represents:

- A. Atoms
- B. Molecules
- C. Particles
- D. All

Q17. Mole fraction sum equals:

- A. 0
- B. 1
- C. 2
- D. Depends

Q18. If pressure doubles, volume becomes:

- A. Half
- B. Double
- C. Same
- D. Zero

Q19. Which has same moles?

- A. 22.4 L O_2
- B. 44 g CO_2
- C. 6.02×10^{23} particles

D. All

Q20. Gram equivalent =:

A. Mass/equiv wt

B. Mass $\times$ equiv wt

C. Mole $\times$ mass

D. None

Answer Key

Q1: B
Q2: A
Q3: A
Q4: A
Q5: B
Q6: A
Q7: A
Q8: A
Q9: B
Q10: A
Q11: A
Q12: A
Q13: A
Q14: A
Q15: A
Q16: D
Q17: B
Q18: A
Q19: D
Q20: A